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Why Do Phase Transitions Matter in Minds?

Abstract: *Subjective experience suggests that we continuously observe, perceive, and evaluate the environment as we make decisions and intentional actions. The percept of continuity of our cognition, however, is an illusion. In the past decades, ample experimental evidence has been accumulated indicating that cognition evolves through a sequence of discontinuities and transients, and there are discernable neural processes correlating with the cognitive sequences. These discontinuities are crucial in the intentional action–perception cycle, as they mark the cognitive ‘aha’ moment of deep understanding and conscious insight. Walter Freeman developed a hierarchical model to describe these physiological findings, culminating in the cinematic theory of cognition. Following the traditions of meta-stability, transient neurodynamics, and brain chaos, discontinuities are described as phase transitions in the cortical neuropil. This approach not only leads to deep theoretical insights of neural correlates of cognition and consciousness, but it also helps to form a conceptual framework to create artificially intelligent devices that are in harmony with humans and serve peaceful development of humanity.*

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1. Introduction

Based on his immense multidisciplinary background, Walter Freeman's legacy covers a wide range of research fields, including neurophysiology; higher-level cognitive functions, especially in mammals; computational, mathematical, and physical models of neural and cognitive functions; and philosophical and ethical issues of brain research. Here we focus on his work regarding the interpretation of the experimental findings on the self-organized sequence of neural and cognitive frames, also called the cinematic theory of cognition (Freeman, 2007), and its broader philosophical and ethical implications. It has been almost two decades since the publication of the work by Freeman in this journal on 'Restoring to cognition the forgotten primacy of action, intention and emotion' (Freeman, 1999), in which he points out the limitations of cognitivism and representationalism, and the constraints posed by reductionism in today's complex landscape of cognitive science when trying to understand the basic tenets of natural and artificial intelligence and consciousness. Freeman outlines a systemic approach to mind, soul, intentionality, and values-based decision making with the potential for a better and sounder scientific paradigm of consciousness.

Representationalism views intelligent behaviour as symbolic manipulation of mental representations according to specific logical rules (Minsky and Papert, 1969; Minsky, 1974). Traditional artificial intelligence (AI) produced spectacular successes; see, for example, Newell and Simon (1988). Following their prominence in modelling intelligence for decades, it is natural that neuroscientists try to identify symbolic representations in brains. There are various supporting arguments for localized representations, such as the concept of grandmother cells (Lettvin, 1995), and single neuron recordings indicate the presence of dedicated neurons responding to specific input stimuli during cognitive tasks such as, for example, face recognition (Quiroga *et al.*, 2002). As a result, intracranial recordings in human and animal neocortices are widely cited to lend experimental support to representationalism. There is a caveat in the interpretation of these single neuron experiments, however. Were grandmother neurons randomly distributed in the neocortex, the likelihood of finding any of them would be vanishingly small. It would be practically impossible to identify the visual image which would evoke a measurable response in a given probe by flashing even hundreds of visual images in front of the patient. The fact that such responsive cells can be found at all

implies that they are concentrated in neural assemblies distributed in the neocortex. Potential candidates are Hebbian cell assemblies that are formed by learning to enact a designated task (Freeman and Quiroga, 2012). Once any part of a Hebbian cell assembly is stimulated, the whole assembly responds, which may contribute to the experimentally observed effect.

As an alternative to the representationalist approach to brain dynamics and intelligence, following Merleau-Ponty's traditions, Dreyfus proposes the conceptual framework of situated intelligence (Dreyfus, 1993; 2007). Situated intelligence ascertains that intelligence is defined in the context of the environment; therefore, a preset and fixed symbol system, whatever complex it might be, cannot grasp the essence of intelligence. Pragmatic implementations of situated intelligence and embodiment (Thompson and Varela, 2001) find their successful applications in embodied intelligence and robotics (Brooks, 1988; Barsalou, 2008). Dreyfus embraces Freeman's dynamical systems approach to brains as a possible path towards understanding human intelligence and building intelligent devices. Freeman's neurodynamics describes the brain state through a trajectory across a high-dimensional space (Freeman, 1975; 2008b). The trajectory evolves through a sequence of metastable states (Freeman and Quiroga, 2012; Kozma and Freeman, 2016). These metastable states may be viewed as intermittent 'symbols'; however, they are transient and disintegrate soon after they emerge (Kozma and Freeman, 2008; 2013). The corresponding dynamic view has been crystallized in the concept of the cinematic theory of cognition (Freeman, 2007). According to the cinematic theory, cognition is not a smooth, continuous process in time; rather it is a sequence of metastable cognitive states, which can be viewed as movie frames. Such frames exist for about 100–200 ms, and then they briefly collapse. The collapse of the states signifies the shutter and it takes about 10–20 ms. The quasi-periodic sequence of the frames (metastable patterns) is the movie of the brain as cognitive processing evolves in time (Kozma and Freeman, 2017).

There are various ways to model the experimental findings on rapid changes in brain dynamics, including chaotic itinerancy (Tsuda, 2001); mesoscopic chaotic neurodynamics (Liljenström, 2010); heteroclinic channels (Rabinovich, Sokolov and Kozma, 2014); and behavioural manifestations of metastability as described by the complementarity principle (Kelso and Engstrom, 2006; Tognoli and Kelso, 2015). Sudden transitions from ordered states (AM patterns) to disorder (collapse of patterns) can be described using the concept of

phase transitions. In many influential approaches to cognition, phase transitions are postulated as key manifestations of conscious experience. Quantum field theory (QFT) provides a powerful tool to describe phase transitions between coherent and not-coherent states in the physical world. QFT has been employed to describe dissipative brain dynamics in explaining cognitive processing (Freeman and Vitiello, 2006; Freeman, Kozma and Vitiello, 2012; Freeman *et al.*, 2015).

In an alternative approach, random graph theory and percolation processes are employed to describe phase transitions in large-scale cortical networks, and to interpret experimental findings on the discontinuities between metastable states. Neuropercolation is a family of probabilistic models based on the mathematical theory of probabilistic cellular automata over lattices and random graphs and it is motivated by structural and dynamical properties of the cortical neuropil (Kozma *et al.*, 2005). Neuropercolation extends the concept of phase transitions to large interactive populations of nerve cells as an alternative to models based on differential equations. The neuropercolation model describes various dynamical properties of metastable activity patterns in the cortical neuropil, and it implements principles of intentional neurodynamics (Kozma, Puljic and Freeman, 2013; Kozma and Puljic, 2015).

We expand on Freeman's cinematic theory of cognition with an emphasis on how the brain creates knowledge and meaning from disembodied information (Freeman, 2004a; Kozma and Freeman, 2017). We point to the crucial role of the 'aha' moment of deep insight and understanding, as the manifestation of discontinuity and phase transition in the brain. Our work is based on EEG and ECoG data, specifically rabbit ECoG (Barrie, Freeman and Lenhart, 1996). These studies reveal that brain activity patterns are intermittent both in time and space and give rise to metastable amplitude modulated (AM) neural patterns. Metastable AM patterns exist for relatively long time periods (fractions of a second), while the transition from one state to the other is fast (in the order of several milliseconds). We employ pragmatic information for the identification of neural correlates of cognition and consciousness (Kozma, Davis and Freeman, 2012; Davis, Gillett and Kozma, 2015). Following a phase transition event with singular space-time dynamics, structure and order emerges in the form of metastable spatial amplitude modulated (AM) patterns observed in brain signals (Kozma and Freeman, 2013). AM patterns show that the creation of knowledge and meaning requires significant

energy consumed by the brain as an open thermodynamic system converting energy and information into meaning and knowledge (Friston, 2009). This analysis provides a constructive approach to reconcile the contradictions inherent in perceived mental representations and the immense complexity and flexibility of brain dynamics. It may help to establish a harmony between the everlasting human desire to create tangible chunks of knowledge reflecting our limited grasp of reality and the infinite complexity of the world in which our body and brain reside. In line with Freeman's legacy, this work aims at defining a conceptual framework to serve the sustainable and peaceful development of mankind.

2. Review of Basic Principles of Freeman Neurodynamics

During the individual's lifetime, human beings are presented with innumerable repetitions of different kinds of intentions and moments of enacting intentional arcs; this is manifested in Freeman's intentional neurodynamics (Freeman, 2008a). The brain has to continuously actualize, based on past memories and the projections of future memories, the network of knowledge and meaning which is necessary to infer or even have insights into the courses of action which optimizes his or her actions when faced with the decisions to be made in present time (Werbos, 2007). A systematic implementation of Freeman's principles of intentional neurodynamics employs a hierarchical approach based on Freeman K (Katchalsky) sets (Freeman, 1975; Kozma and Freeman, 2008), named after Aharon Katzir Katchalsky, an early pioneer of spatio-temporal neural dynamics. The hierarchy includes microscopic, mesoscopic, and macroscopic scales, through the sequence of K0, KI, KII, KIII, KIV, and KV sets.

Here we list the ten principles of Freeman neurodynamics (Freeman, 2015):

1. State transition of an excitatory population from a point attractor with zero activity to a non-zero point attractor with steady-state activity by positive feedback (KI).
2. Emergence of oscillations through negative feedback between excitatory and inhibitory neural populations (KII). Control of beta-gamma carrier frequency range achieved in KII with low feedback gain.

3. State transitions from a point attractor to a limit cycle attractor that regulates steady-state oscillation of a mixed excitatory–inhibitory cortical population (KII). It is achieved in KII with sufficiently high feedback gain.
4. Genesis of chaos as background activity by combined negative and positive feedback among three or more mixed excitatory–inhibitory populations (KIII). Achieved by coupling KII oscillators into a KIII unit with incommensurate frequencies.
5. Distributed wave of chaotic activity that carries a spatial pattern of amplitude modulation (AM) made by the local heights of the wave (KIII). These mark the emergence of AM patterns in KIII.
6. Input-driven increase of the nonlinear feedback gain, which results in the modulation of an amplitude modulation pattern as the first step in perception.
7. The embodiment of meaning in AM patterns of neural activity, which are shaped by synaptic interactions that have been modified through learning. This is the manifestation of Hebbian learning in KIII layers.
8. Attenuation of microscopic sensory-driven noise and enhancement of macroscopic amplitude modulation patterns by divergent–convergent cortical projections. The intracortical connections lead to the interaction of KIII sets and the formation of KIV.
9. Gestalt formation and prefference through the convergence of external and internal sensory signals leading to the activation of the attractor landscapes, and leading to intentional action. Emergence of attractor landscape and intentional neurodynamics in KIV.
10. Global integration of AM frames at the theta rates through neocortical phase transitions representing high-level cognitive activity of the hemisphere in the KV model.

The fundamental basic building block of Freeman neurodynamics is postulated as the K0 set, which comprises the local population encompassing up to 10,000 neurons, that produces an input–output mapping expressed by the asymmetric sigmoid function (Freeman, 1975). The original K models used a system of second-order ordinary differential equations (ODEs) with distributed parameters to describe K0 dynamics. In more recent manifestations of the K model, the neuropercolation approach has been implemented (Kozma *et al.*, 2005; Harter and Kozma, 2005), which uses mathematical results of

percolation theory to describe activity propagation across the cortical neuropil. Neuropercolation had some advantages as compared to ODEs in describing sudden changes, discontinuities in the spatio-temporal dynamics (Kozma *et al.*, 2005; Kozma and Puljic, 2015).

Freeman and colleagues extensively developed KI, KII, and KIII sets during the 70s, 80s, and 90s, which not only provided tools to interpret experimental neurophysiology findings, but also build novel highly recurrent neural network architectures capable of solving very difficult classification problems and other challenges posed to AI research (Kozma, 2007). KIV models of intentional neurodynamics have been developed since the early 2000s, to model multisensory fusion, Gestalt formation, and decision making in animals and man-made devices (Kozma *et al.*, 2007; Kozma and Freeman, 2009). KV sets describe large-scale, 6-layer mammalian neocortical networks (Freeman, 2004a). In each hemisphere the neocortex is a unified organ maintaining meta-stable dynamics near critical states. Criticality allows the system react to changes very rapidly, through phase transitions, which is a singularity when local sensory inputs are converted into global patterns of cortical activity in an instant, within a few milliseconds. They correspond to high-level cognition in humans, such as language and reasoning, and they represent an ongoing research field (Kozma and Freeman, 2009; 2016).

3. The Cycle of the Creation of Knowledge and Meaning

3.1. Pragmatic Information to Describe Changes in an Order Parameter

Here we first characterize key aspects of pragmatic information related to intention, decision making, and action, followed by the rigorous statistical description of pragmatic information indices. A tennis match provides a good example of the significance of pragmatic information, when one player derives information from the movements of the other in order to anticipate his or her actions. Pragmatic information is associated with several events: (1) the spoken word that carries meaning for action; (2) information associated with body language even when nothing is verbalized; (3) stress hormone levels or amplitude wave patterns as measured from brain oscillations or wave-like patterns (biological markers). It could be easily conjectured that pragmatic information emerges hand in hand with the creation of

meaning together with intention, before even initiating an action such as shouting at or knocking on my neighbour's door to invite him for dinner. In recent years, scientists in different fields, in particular in physics, have used pragmatic information to show that this concept is relevant in describing self-organizing systems based on approaches like synergistic and nonequilibrium thermodynamics.

Atmanspacher and Scheingraber (1990) have given a thorough explanation of pragmatic information following semiotic traditions, where information has been considered from three different perspectives as follows: syntactics, semantics, and pragmatics. Semantic information is derived from grammar and linguistic ingredients together with the articulation of sounds. The deep connection between pragmatic information and meaning is the key aspect that distinguishes it from pure semantic information. Here we focus on the power and value of pragmatic information, which is associated with an urge for action, that is in principle identifiable in preparatory body movement, or even in brain dynamics (urge or preparation for action) that precedes body movement. When we say to another friend, 'I am going for a bike ride', we are revealing our intention for that action via semantic information. In contrast, if I quietly grab my bike and start to ride in front of my friend while waving, I also reveal the same intention, however via pragmatic information. If I could measure brain waves associated with the intention to grab my bike and clearly identify that kind of brain pattern previous to the physical action, then, that would also be considered pragmatic information.

Shannon and Weaver (1963) showed us how to measure information transferred via a communication channel while guaranteeing, to some extent, the semantic integrity of the message that travels from sender to receiver. The widely used Shannon information index, however, lacks the power to provide for the information content of the message much less the meaning that the message carries for action. Because of that, we are faced with the need to appeal to other kinds of indices more suited for that task. Living organisms like human beings may read a paragraph from a text and when the brain processes the information this may trigger a set of actions, which physically means work, when potential energy is converted into kinetic energy.

Freeman's interest in pragmatic information for characterizing intentional neurodynamics processes was inspired by Atmanspacher and Scheingraber (1990), and the present work is a natural extension of his groundbreaking initial steps. We use some generalized pragmatic information indices to measure the level of order or structure in

windows (frames) of brain dynamics associated with the meaning of different stimuli in contrast with the level of disorder in background activity. In close collaboration with Walter Freeman in recent years, we held many conversations and work sessions that lead us to the conclusion that we had to develop a methodology for brain dynamics movies, which provides us with robust quantitative indices to measure the distinctions that we saw in qualitative cinematic brain patterns (Kozma, Davis and Freeman, 2012; Davis, Gillett and Kozma, 2015). This methodology included pragmatic information indices, since we anticipated that they are able to capture nonlinear changes in brain dynamics, which are associated with the creation of knowledge and meaning based on experimental designs that Freeman carried out in his neurophysiology lab at UC Berkeley (Barrie, Freeman and Lenhart, 1996).

We explore pragmatic information to translate the concept of meaning into the language of brain dynamics. The proposed pragmatic information index ($H(t)$) is the ratio between an ‘order parameter’ and the rate of change of the order parameter in time. For example, consider a multidimensional signal vector $A(t)$. In this case, the order parameter can be an amplitude measure, such as the squared amplitude $\|A(t)\|^2$, and the rate of change of the order parameter can become $D(t) = \text{dist}(\|A(t+1)\|^2, \|A(t)\|^2)$, thus $H(t) = \|A(t)\|^2/D(t)$ (Freeman, 2004b; Atmanspacher and Scheingraber, 1990). This ratio represents the energy transactions reflected in electrical oscillations measured in the cortex or the scalp, which would be also reflective of the meaning and the knowledge encoded in brain signals. We explore various variants of the basic pragmatic information index, depending on the interpretation of the signal magnitude (energy); the rate of change can be determined using Euclidean distance or other suitably selected distance measures.

3.2. Implementation of Pragmatic Information Indices in Experimental Analysis

Data were collected using ECoG (Barrie, Freeman and Lenhart, 1996) on rabbit cortices, as well as scalp EEG experiments on human beings both in laboratory conditions and in a natural setting (Davis, Gillett and Kozma, 2015; Davis *et al.*, 2017). Information has been extracted on the activity of the brain over multiple frequency bands in response to various stimuli. In the case of human participants, various technologies were tested, where brain areas of different sizes were covered,

sometimes using a very small spacing between electrodes, while in other experiments the spacing between electrodes was relatively larger. In experiments with higher spatial resolution, it is expected that the evolution of the signal in time would provide images and data well suited for spatio-temporal analysis. Here we focus on results obtained with rabbit ECoG analysis in Freeman's neurophysiology lab at UC Berkeley (Barrie, Freeman and Lenhart, 1996; Freeman *et al.*, 2000).

Rabbit ECoG data have been obtained via electrode arrays placed over sensory cortices of rabbits (visual, auditory, olfactory, and somatosensory cortices) using 8 x 8 arrays, with electrode spacing of about 1 mm (Barrie, Freeman and Lenhart, 1996). A single trial consists of a 6 s recording period, with a stimulus conveyed briefly after 3 s. Thus the first 3 s correspond to pre-stimulus conditions, while the other 3 s is post-stimulus state. The 6 s trials were repeated many times for two different conditions, CS+ and CS-, i.e. positive and negative reinforcements, respectively. The pre-stimulus period describes background activity while the post-stimulus period starts with a 1 s window, where knowledge and meaning are presumably created. There are moments of very high synchronization within 0.25–0.5 s following the stimuli, which relate to intense attention. This is followed by a learning and integration period with desynchronization, at about 0.5–1 s after the stimulus. After the 1 s post-stimulus period, the trial continues for another 2 s, during which the signals return to background activity.

We structured the methodology to study various frequency bands and also make comparison between frequencies. We employed Hilbert analysis, which was introduced earlier to monitor rapid changes in the measured signals (Freeman *et al.*, 2000; 2003; Stam, Nolte and Daffertshofer, 2007). First we decide which band to study, and what bandwidth to use, and filter the signal over the specified band. Then the band-passed signal is Hilbert transformed to obtain a set of signals: analytic amplitude (*AA*), analytic phase (*AP*), together with the original signal amplitude (*SA*) without Hilbert transform. For *AA*, *AP*, and *SA*, we compute a set of amplitude measures, such as range of analytic amplitude (*RAA*), range of squared amplitude (*RSA*), and the squared analytic amplitude (*SAA*), for a sequence of time windows. These are used as the amplitude measures for the order parameter $\|A(t)\|^2$, according to the computations of the pragmatic information $H(t)$ as described in the previous section. This approach leads to a family of pragmatic information indices, *HRAA*, *HRSa*, and *HSAA*, respectively. There are additional possible variations regarding the

way the distance measures are calculated in the denominator of $H(t)$ (Davis and Kozma, 2012a,b; Davis, Gillett and Kozma, 2015). With these pragmatic information indices we are equipped to perform both visual and statistical analysis to uncover the stages or steps in the creation of knowledge and meaning, since these indices are able to capture changes in brain dynamics associated with different brain states or stimuli processing.

3.3. *The Cycle of Creation of Knowledge and Meaning (CKM Cycle)*

Here we introduce the cycle of the creation of knowledge and meaning, which we call the CKM cycle for brevity. The CKM cycle has been identified and thoroughly analysed based on the evaluation of ECoG data (Kozma, Davis and Freeman, 2012). Here we summarize the results of the ECoG analysis. It was found that a very high synchronization was associated with low analytic amplitudes (AA) and a very dramatic desynchronization was coupled with a dramatic drop in amplitude. It was conjectured that the spatially distributed AA patterns, together with the original signal amplitude (SA), might carry useful information concerning the meaning that the animal associates to the stimuli and its relevance in decision making.

Our results show that a sequence of five characteristic steps can be distinguished in the ECoG spatio-temporal dynamics during the 1 s window of time following the presentation of the stimulus. The sequence of five steps describes how synchronization–desynchronization effects with sudden transitions in spatio-temporal neurodynamics can be relevant to cognition and awareness. They are described as follows:

- Step 1 (0–0.1 s): Initial impression, which may be termed the ‘awe’ (or noticing) moment.
- Step 2 (0.1–0.3 s): Chaotic exploration of memory traces with highly distributed and desynchronized patterns; search for meaning.
- Step 3 (0.3–0.45 s): Recognition/identification of the searched clue/decision and it can be termed the ‘aha’ moment.
- Step 4 (0.45–0.6 s): Integration of the new knowledge in a chaotic process and modifying the dynamic landscape of cognition.
- Step 5 (0.6–0.9 s): Return to the basal, background/default level of toward the end of the 1 s post-stimulus period.

Steps	Cycle of Creation of Knowledge & Meaning (CKM)	Characterization Using Pragmatic Information	Freeman Action–Perception Cycle
1	<u>Awe</u> (0–0.1 s) <i>Noticing</i>	<u>Low Pragmatic Information</u> Low $ A(t) ^2$ High synchrony	<u>Raw sense-data</u> Anticipate the consequences of action Select and transduce stimuli Sensory perception Hebbian assemblies Accommodation
2	<u>Chaotic Exploration</u> (0.1–0.3 s) <i>Search for meaning</i>	<u>Increased Pragmatic Information</u> Strong rise of $ A(t) ^2$ with dramatic drops (null spikes) Desynchronization	<u>Sensory cortex</u> Explore attractors & AM patterns Adapt, use knowledge <u>Limbic system</u> Multisensory percept, Gestalt
3	<u>Aha</u> (0.3–0.45 s) <i>Cognitive insight, recognition, decision</i>	<u>Drop of Pragmatic Information</u> Drop of $ A(t) ^2$ Tendency to synchronization	<u>Neocortex</u> Global wave packet <i>In Humans</i> : Speech, symbolic cognition, decision
4	<u>Chaotic Integration</u> (0.45–0.6 s) <i>Knowledge integration</i>	<u>High Pragmatic Information</u> Rise of $ A(t) ^2$ with abundant null spikes Strong desynchronization	<u>Sensory cortex</u> Adaptation of attractors, knowledge, AM patterns <u>Limbic system</u> Completing action–perception cycle and Gestalt formation
5	<u>Return to Background</u> (0.6–0.9 s) <i>Set to basal activity level</i>	<u>Decreased Pragmatic Information</u> Low $ A(t) ^2$ Synchronization with Periods of desynchronization & high $ A(t) ^2$	<u>All Cortices</u> Maintain the changed/new state Expectative background dynamics Relax into habitual brain activity

Table 1. Cycle of creation of knowledge and meaning creation and Freeman intentional action–perception cycle (based on Davis, Gillett and Kozma, 2015).

Outside the 1 s post-stimulus window, the system in general shows good synchronization between channels with low amplitudes. This is accompanied by periods of mild desynchronization showing relatively higher amplitudes. Overall, we observe clear differences between the CS+ and CS− collective responses.

The observed five steps comprise the postulated cycle of CKM. In order to illustrate our findings, we produced a diverse set of movies showing the spatio-temporal patterns of *AA*, *AP*, and *SA*, and other statistical features (Davis, Gillett and Kozma, 2015). After watching these movies over and over again, and introducing classification methods, discriminant analysis, and clustering, we accumulated statistical evidence of the cycle of CKM and the emergence of awareness experience in the neocortex (Davis and Kozma, 2013).

It is important to point out that experimental studies in various laboratories indicate epochs in sensory processing related to the five cognitive stages mentioned above. The rabbit data shown here involve classically conditioned behaviour, but operant tasks might expand or contract states or maybe add more, and different sensory systems might slow down or speed up the states. For example, the work of Frederick *et al.* (2016) shows that there are at least two epochs during odour sampling. These include an early gamma-based processing with a dominantly feed-forward processing state, and beta oscillations after about 0.2 s, which engage the entire system as the animal prepares for the response (*ibid.*). Other work shows that the gustatory system goes through sudden state changes within a taste-sampling period (Sadacca *et al.*, 2016).

4. Interpretation of Experimental Findings as Manifestations of the Cycle of Knowledge and Meaning Creation

Pragmatic information (PI) indices provide crucial support to quantify the steps of the CKM cycle (Davis, Gillett and Kozma, 2015). Here we summarize the main findings and also link these results to Freeman's work on the intentional action–perception cycle. In general, all PI indices show significant increase during the period 0.2–1.0 s following the stimulus. This means that these indices are able to filter out the initial strong impact of the sensory stimulus during the 'awe' period (step 1), caused by the imprint of sensory data through the sensory tracts on the primary sensory cortex. This imprint has been called Type I response and it does not involve higher cognitive

functions (Kozma and Freeman, 2002). There are two major peaks in the 0.2–0.8 s period. The details of these effects are studied using frequency bands alpha, beta, and gamma (Davis, Gillett and Kozma, 2015). It is important to select a suitable time resolution for determining the PI indices. A time resolution of 1/16 s is close to the optimum window size for the gamma band indicated by Freeman (2008b). The results in Davis, Gillett and Kozma (2015) use a somewhat lower time resolution (1/8 s), which is sufficient to describe the hypothesized cycle of creation of knowledge and meaning.

Table 1 summarizes the main observations regarding the characterization of the CKM cycle using PI indices and connects these results with the Freeman action–perception cycle (Freeman, 2008a; Davis, Gillett and Kozma, 2015). Step 1 (awe) shows hyper-synchronization, followed by desynchronization at step 2 (chaotic exploration). Step 2 also demonstrates an increased amplitude measure, except for some dramatic drops in the amplitudes due to null spikes. Null spikes arise at some locations over the cortical sheet when the amplitude measure drops several orders of magnitude in a narrow window of time (of a few ms). Overall, the PI index increases at step 2. At the aha moment (step 3), the synchronization increases, the amplitude measure decreases, and there is a clear decrease in the PI value. Step 4 (chaotic integration) shows similar behaviour to step 2 with high amplitudes (except some null spikes) and marked desynchronization, while PI is high. The high PI values at steps 2 and 4 provide a practical way to locate the aha moment (step 3) with lower PI. Finally, at step 5, synchronization increases, the amplitudes stabilize and PI decreases and the overall activity returns to a basal level.

Freeman's views on knowledge, meaning, and intentionality provide the neurophysiological underpinning for the systematic evaluation of the CKM cycle using pragmatic indices (Freeman, 2008b). Column 4 in Table 1 displays components of Freeman's action–perception cycle. This includes anticipation, sensory perception, activation of the amplitude modulation (AM) patterns in sensory cortices, and multisensory percept formation (Gestalt) in the limbic system. The aha moment (step 3) plays a special role in the cycle of CKM, manifested in the sudden emergence of cognitive insight and recognition; this is emphasized by the grey outline of row 3 in Table 1. The aha moment is a singularity in the space-time dynamics, when the amplitude measure of the order parameter vanishes and the rate of change explodes. Freeman calls this the 'global wave packet', which rapidly grows to cover the whole hemisphere. The emergence of the global

wave packet manifests a condensation process, which is a phase transition from high-dimensional chaotic dynamics (disorder) to low-dimensional order. Thorough analysis showed that this emergent process is initiated at some incipient phase cone in the neocortical sheet. The initial phase cone grows in size and produces a propagating phase gradient. Once the phase cone reaches a size comparable to the hemisphere, it disintegrates and the conditions are formed for the emergence of a new AM pattern over the sensory cortices, the intentional action–perception cycle is complete, and the dynamics returns to the basal state.

Step 3 (aha moment) is crucial for cognitive processing and also for conscious experience. As step 3 is a space-time singularity in the cortical neurodynamics, the concept of phase transition is indeed crucial to describe consciousness. We support this argument by invoking Baars' global workspace theory of consciousness. Namely, we introduce the hypothesis that the rapid propagation of phase dispersion over the hemisphere is the manifestation of the cognitive broadcast as described in Baars' global workspace theory (Baars, 1997). In the hierarchical model of neurodynamics using Freeman K-sets, we observe intermittent desynchronization over large cortical areas for a short period. This is the period of phase transition, when large phase gradients travel rapidly across the cortex. This rapid propagation of phase synchronization–desynchronization resembles the rapid propagation of conscious broadcast as described by the global workspace theory. Our ECoG experimental studies using PI indices indicate that a large-scale cortical desynchronization followed by rapid synchronization and the collapse of analytic amplitudes are associated with the 'aha' effect of cognitive processing. In conclusion, this leads us to the hypothesis that cortical singularities and phase dispersion could serve as neural correlates of the conscious broadcast.

5. Conclusion

Subjective experience suggests that we continuously observe, perceive, evaluate the environment, and make conscious decisions and intentional actions. However, the percept of continuity of our cognition is an illusion (Kozma, 2013). Our brains create the conscious percept of continuity to help our mind to deal with the complex, discontinuous sensing (Mercier and Sperber, 2011). In the past decades, ample experimental evidence has been accumulated indicating that cognition manifests a sequence of rapid transients, and

there are discernable discontinuous neural processes correlating with the cognitive sequence. The main thesis we put forward here is that discontinuities are not simply important features of neurodynamics, but they are the centrepiece of higher cognition and consciousness, they represent the cognitive ‘aha’ moment of sudden insight and deep understanding.

We propose to study phase transitions in the cortex using pragmatic information (PI) indices in the framework of the cycle of the creation of knowledge and meaning (CKM). This work is closely related to Freeman’s intentional action–perception cycle describing metastable AM patterns and the propagation of global wave packets during conscious experiences. We put forward an hypothesis on the link between cortical phase transitions and Baars’ global workspace theory. In order to grasp the implication of the CKM cycle on conscious and wise decision making based on intentional acts, we have to understand information as an active principle capable of putting in motion different systems, both mechanical and biological.

In our quest to comprehend consciousness and to understand how meaning and knowledge are manifested in the brain, we foresee future studies to measure EEG activity in humans while in states of relaxation and peaceful meditation, accompanied by psycho-physiological coherence mediated by breathing. We also foresee the need to contrast such states with other types of mental activity that manifest in brain dynamics while humans are engaged in unpleasant energy depletion activities leading to stress. These ideals come with a set of questions to be explored and answered in relation to human awareness and consciousness and their relation to spiritual experience and mental health, leading to a better and more intimate cognitive attunement to the human life-world, towards a more inclusive or universal framework of action expressed in social behaviour that can be studied under new emerging paradigms (Davis, 2009). Some of these questions are:

- Could it be that the brief period of desynchronization followed by intense synchronization and intermittency, then leading to the integration of experience and perception into an actualized amplitude pattern, are at the core of what we experience as eureka or ‘aha’ moments when engaged in scientific research, philosophical enquiry, or other types of creative activities?
- Could it be that the vastly reported enlightenment moments associated with religious, spiritual, or mystical experiences fall under the same structure of meaning and knowledge creation?

- Is it possible that people could learn how to function in different societies according to a universal framework of ethics, and overcome unhealthy habits, addictions, disease, and compulsive survival needs?

Following Freeman's footsteps, our goal is to develop tools to improve our understanding of brain dynamics, in particular the cycle of knowledge and meaning creation. This will bring us closer to understanding what human consciousness really is. To answer these questions, we are convinced, is important for the future of humanity and to create conditions for harmonious living in society.

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